Individual Reflection
Medical Question Answering on PubMedQA
EMATM0067 – Introduction to AI and Text Analytics
Group 16

Abhigyan Kashyap
wm25342@bristol.ac.uk
University of Bristol

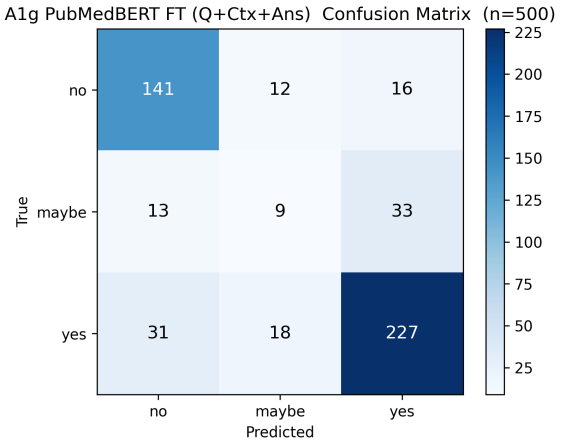

Figure 1: A1g PubMedBERT FT (Q+Ctx+Ans) confusion matrix on the 500-sample held-out test set. 47 yes↔no reversals; only 9 of 55 true *maybe* cases recovered.

1 Contributions and Activity

I oversaw the error-analysis and result-verification track for Group 16 across the duration of the PubMedQA project. My contributions covered the confusion-matrix audit of the representative model (A1g PubMedBERT FT, Q+Ctx+Ans) and of every minority-class variant (M1–M5), the catalogue of misclassified examples by pattern (yes→no, no→yes, maybe→yes, maybe→no) that became Table 5 of the group report, the McNemar significance tests, the pairwise Jaccard error-overlap heat-map, and the K-Means cluster-based per-cluster F1 breakdown. I also implemented a standalone verification script in the code directory that re-derives the headline test-set numbers from the saved A1g predictions and a freshly trained baseline, so anyone on the team can cross-verify the reported numbers without having to retrain the transformers. I tracked tasks on Jira and pushed all artifacts through the shared GitHub repository. Figure 2 shows my GitHub contributions.

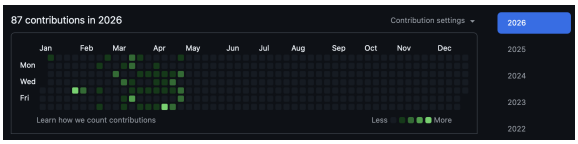

Figure 2: GitHub contribution graph for my account: 87 contributions in 2026, concentrated in the eight-week project window (Feb–May).

2 Technical Understanding

The technical challenge I worked on most closely was diagnosing what kind of error A1g was making and using that diagnosis to guide the team's minority-class interventions. The headline macro-F1 (0.604) hides two qualitatively different failure modes: yes↔no reversals (direction-of-evidence errors, highly relevant in a clinical context) and maybe-class suppression. The A1g confusion matrix in Figure 1 shows 47 yes↔no reversals and, more strikingly, that 46 of 55 true *maybe* cases (83.6%) are forced into yes/no – only 9 are recovered. This separation of errors matters because the two failure modes call for specific fixes: the reversals respond to better evidence representation, whereas maybe-suppression responds to loss reweighting. Figure 3 shows the four representative patterns I catalogued.

In order to validate that A1g is a real lift rather than a lucky split, I ran a paired McNemar's test against the TF-IDF baseline ($b = 164$, $c = 40$, $p < 0.001$) and computed Jaccard error-overlap across the four highest-ranked unique transformer variants. 66 examples (13.2%) were misclassified by all four PubMedBERT variants, implying a hard, shared core that ensembling within the PubMedBERT family will not solve. The cluster-based view highlighted that A1g's per-cluster maybe-F1 ranged from 0.160 (cluster 0) to 0.250 (cluster 2), so semantic regions of the embedding space are unevenly uncertainty-heavy.

Table 5: Representative misclassified examples from A1g (shortened).

Pattern	Question (shortened)	True	Pred
yes→no	Double balloon enteroscopy: efficacious and safe in community?	yes	no
no→yes	Long-term results of transanal vs transabdominal pull-through equal?	no	yes
maybe→yes	30-Day mortality emergency laparotomies: an area of concern?	maybe	yes
maybe→no	HPV and pterygium: virus a risk factor?	maybe	no

Figure 3: Representative misclassified examples I extracted from A1g, grouped by error pattern. Source for Table 5 of the group report.

My analysis helped the team re-categorise the bottleneck: the residual gap was *not* due to model capacity but to the model's limited evidence-sufficiency reasoning. This reframing motivated Tanishk's targeted M1–M5 trade-off (focal loss raising maybe-F1 from 0.192 to 0.237 at the cost of a 0.013 macro-F1 drop), and it shaped Section 3.7 of the report into an error-type analysis rather than another model bake-off.

3 Teamwork

Coordination ran through a Jira board for task tracking, a shared GitHub repository, weekly Teams meetings, a WhatsApp group chat for quick polls and decisions, and Overleaf for the report. My role sat downstream of nearly everyone, so I leaned heavily on clear hand-offs. I worked with Abdullah to verify the 500/500 split was reproducible from random seed 42 with the expected 276/169/55 distribution before running any analysis; with Peidong to compare A1f vs. A1g per-class F1 deltas and isolate the input-vs-backbone contribution; with Zhengjie to diagnose why the calibrated SVM collapsed to predicting *yes*; and with Tanishk to feed back which of the five maybe-class interventions held up on the held-out test set. Abdullah handled data loading, preprocessing, label encoding, splits and 10-fold CV setup, and sanity checks. Peidong built the Axis-1 representation chain: a TF-IDF + LR baseline, frozen BioBERT embeddings with LR, and full BioBERT fine-tuning. Zhengjie ran the classical Axis-2 sweep – a tuned LR, SMOTE-balanced LR, a calibrated SVM, Multinomial NB, and a soft-voting ensemble. Tanishk implemented the advanced transformer and maybe-class models, final integration, tables, figures, and notebook cleanup. I owned the error-analysis layer: confusion-matrix audits, inspection of misclassified examples, cross-model comparison notes, and verification of headline results.